Means Separation and Data Presentation

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice with **means separation**, **linear contrasts**, and **presenting treatment means**.

You will work with an apple yield dataset to practice:

- fitting an ANOVA model
- comparing treatment means using **LSD** and **Tukey** comparisons
- constructing and testing **linear contrasts**
- presenting treatment means in a **figure with grouping letters**

Submit a **knitted PDF/HTML generated from your R Markdown file** to Canvas.

We can start by loading the necessary libraries

```
library(ggplot2)
library(emmeans)
library(multcomp)
library(nlme)
```

Question 1

1 point

Load the dataset `"data/apple.csv"` and display the first few rows of the dataset. This dataset contains apple yield data from an experiment that evaluated foliar treatments. The experiment was conducted as a randomized complete block design with 4 replications.

```
apple <- read.csv('./data/apple.csv')
head(apple)
```

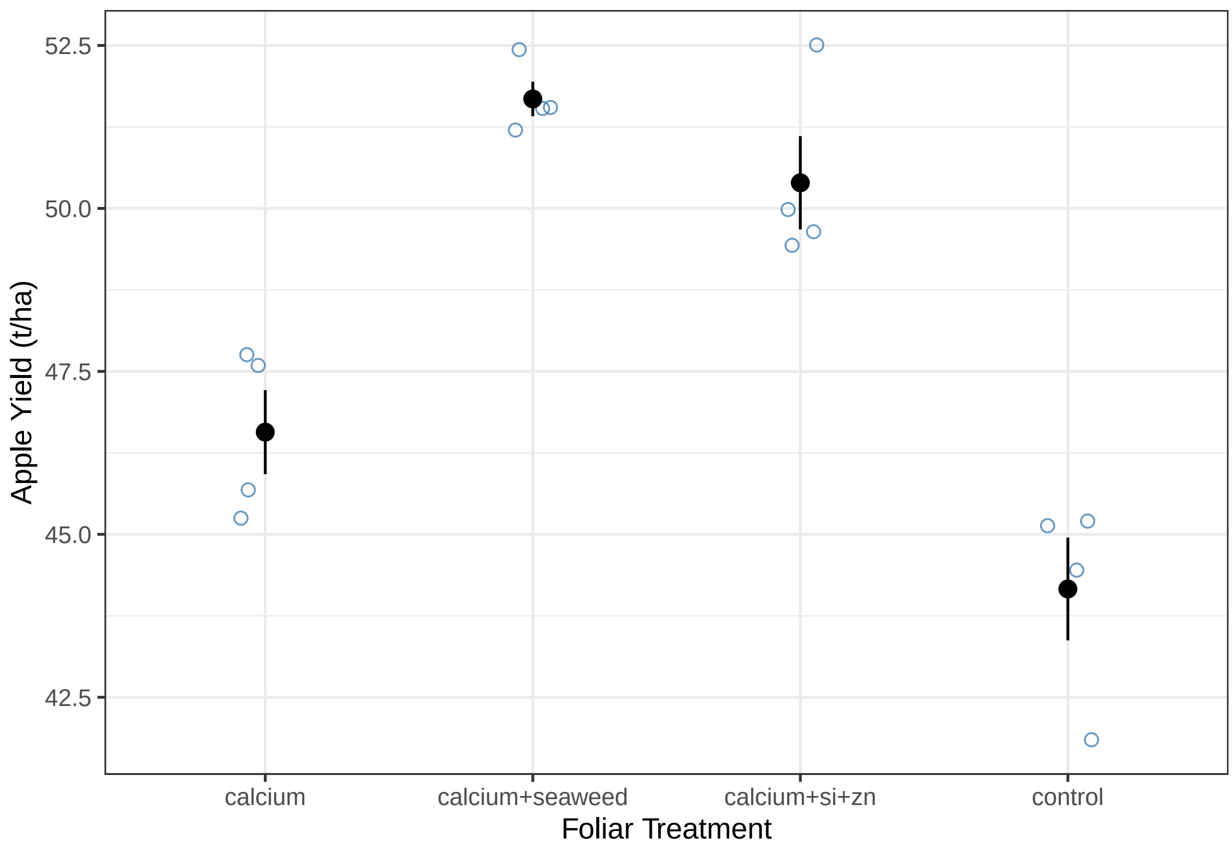
```
##   block      foliar_trt    yield
## 1     1   calcium+si+zn 49.43409
## 2     1 calcium+seaweed 51.55156
## 3     1         calcium 47.75001
## 4     1         control 45.13852
## 5     2   calcium+si+zn 52.51426
## 6     2 calcium+seaweed 52.43407
```

Question 2

2 points

Build a plot to visualize the relationship between `yield` and `foliar_trt`

```
ggplot(apple, aes(x = foliar_trt, y = yield)) +
  geom_jitter(width = 0.1,
              size = 2,
              alpha = 0.8,
              color = "steelblue",
              shape = 1) +
  labs(x = "Foliar Treatment",
       y = "Apple Yield (t/ha)") +
  stat_summary(fun.data = 'mean_se')+
  theme_bw()
```



Question 3

2 points

Fit an ANOVA model testing the effect of `foliar_trt` on apple yield.

Show the ANOVA table.

```r
model <- lme(yield ~ foliar_trt,
             random = ~ 1 | block,
             data = apple)
anova(model)
```

```
##             numDF denDF  F-value p-value
## (Intercept)     1     9 16845.27  <.0001
## foliar_trt      3     9    33.63  <.0001
```

Question 4

2 points

Use the **emmeans** package to compute the estimated marginal means for `foliar_trt`.

Then conduct **pairwise comparisons using LSD inference**.

Show the output.

```r
emm <- emmeans(model, ~ foliar_trt)
pairs(emm, adjust = 'none')
```

```
##  contrast                               estimate    SE df t.ratio p.value
##  calcium - (calcium+seaweed)               -5.11 0.843  9  -6.060  0.0002
##  calcium - (calcium+si+zn)                 -3.83 0.843  9  -4.535  0.0014
##  calcium - control                          2.41 0.843  9   2.853  0.0190
##  (calcium+seaweed) - (calcium+si+zn)        1.29 0.843  9   1.525  0.1616
##  (calcium+seaweed) - control                7.52 0.843  9   8.913 <0.0001
##  (calcium+si+zn) - control                  6.23 0.843  9   7.388 <0.0001
##
## Degrees-of-freedom method: containment
```

Question 5

3 points

Now conduct the same pairwise comparisons using **Tukey's HSD adjustment**.

Show the results and briefly explain how they differ from the LSD-style comparisons.

```r
pairs(emm, adjust = 'tukey')
```

```
##  contrast                               estimate    SE df t.ratio p.value
##  calcium - (calcium+seaweed)               -5.11 0.843  9  -6.060  0.0009
##  calcium - (calcium+si+zn)                 -3.83 0.843  9  -4.535  0.0063
##  calcium - control                          2.41 0.843  9   2.853  0.0749
##  (calcium+seaweed) - (calcium+si+zn)        1.29 0.843  9   1.525  0.4628
##  (calcium+seaweed) - control                7.52 0.843  9   8.913 <0.0001
##  (calcium+si+zn) - control                  6.23 0.843  9   7.388  0.0002
##
## Degrees-of-freedom method: containment
## P value adjustment: tukey method for comparing a family of 4 estimates
```

Question 6

3 points

Now test a **linear contrast** comparing

calcium plus additional ingredients

versus

calcium alone.

First examine the order of treatment levels in `foliar_trt`. Then compute the estimated marginal means using `emmeans`.

Construct the appropriate **contrast coefficients** and test the hypothesis using the `contrast()` function.

Briefly interpret the results.

```
contrast(emm,
         method = list(
           calcium_vs_alone = c(1, -1/2, -1/2, 0)
         ))
```

```
##  contrast          estimate   SE df t.ratio p.value
##  calcium_vs_alone     -4.47 0.73  9  -6.117  0.0002
##
## Degrees-of-freedom method: containment
```

Question 7

1 point

Use `cld()` to obtain **Tukey grouping letters** for the treatment means.

Show the output.

```
cld(emm, adjust = 'tukey', Letters = letters)
```

```
## Note: adjust = "tukey" was changed to "sidak"
## because "tukey" is only appropriate for one set of pairwise comparisons
```

```
##  foliar_trt      emmean    SE df lower.CL upper.CL .group
##  control           44.2 0.636  3     40.8     47.6  a
##  calcium           46.6 0.636  3     43.2     50.0  a
##  calcium+si+zn     50.4 0.636  3     47.0     53.8   b
##  calcium+seaweed   51.7 0.636  3     48.3     55.1   b
##
## Degrees-of-freedom method: containment
## Confidence level used: 0.95
## Conf-level adjustment: sidak method for 4 estimates
## P value adjustment: tukey method for comparing a family of 4 estimates
## significance level used: alpha = 0.05
## NOTE: If two or more means share the same grouping symbol,
##       then we cannot show them to be different.
##       But we also did not show them to be the same.
```

Question 8

1 point

Create a **plot using ggplot2** that shows:

- treatment means
- confidence intervals
- grouping letters above the bars

Your final figure should resemble the type of plot shown in the lecture.